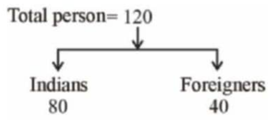


Considering that,
Any person can include
One programme or more than two programme on all programme.
$\therefore$ The sum of active person of all programme can not be 30.
$\because$ Total villagers = 30
$\therefore$ The number of villagers whose age under 25 = 24
and the number of villagers who are literate = 10
Hence,
It is clearly said that, some villagers under 25 years of age are literate.

64. In a group of 120 persons, 80 are Indians and rest are foreigners. Further, 70 persons in the group can speak English. The number of Indians who can speak English is
(a) 20 (b) 30
(c) 30 or less (d) 30 or more

Ans. (d) :



$\because$ 70 persons in the group can speak English.
If all 40 foreigners speak English, then 30 Indians will speak English.
If 39 foreigner speak English, then 31 Indians will speak English.
" " " " " "
" " " " " "
" " " " " "
If 1 foreigner speaks English, then 69 Indians will speak English.
If No one foreigner speaks English, then 70 Indians will English
Hence, the number of Indians who can speak English is 30 or more.

65. Consider all 3-digit numbers (without repetition of digits) obtained using three non-zero digits which are multiples of 3. Let S be their sum.
Which of the following is/are correct?
1. S is always divisible by 74.
2. S is always divisible by 9.

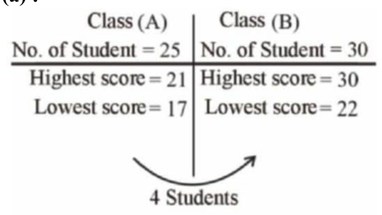
Select the correct answer using the code given below
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : According to question,
Required number 123, 231, 321
126, 261, 612
129, 291, 912
Similarly all numbers obtained
$\because$ the numbers are without repetition of digits and non zero.
S = (666) + (995) + …….
Which is divisible 3 and 37.
Hence, S is always divisible by 74 and S is always divisible by 9.

66. There are two Classes A and B having 25 and 30 students respectively. In Class-A the highest score is 21 and lowest score is 17. In Class-B the highest score is 30 and lowest score is 22. Four students are shifted from Class-A to Class-B.
Consider the following statements :
1. The average score of Class-B will definitely decrease.
2. The average score of Class-A will definitely increase.

Which of the above statements is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) :



If a Student who obtained highest score is shifted from class A to class B, then his position in class B will be lower than the minimum marks of class B.
Hence, the average score of class B will definitely decrease.

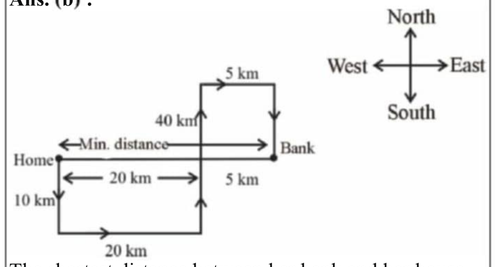
67. A student appeared in 6 papers. The maximum marks are the same for each paper. His marks in these papers are in the proportion of 5 : 6 : 7 : 8 : 9 : 10. Overall he scored 60%. In how many number of papers did he score less than 60% of the maximum marks?
(a) 2 (b) 3
(c) 4 (d) 5

Ans. (b) : Let, obtained marks by the student in 6 papers
= 5x, 6x, 7x, 8x, 9x, 10x,
Total marks = 5x + 6x + 7x + 8x + 9x + 10x $\Rightarrow$ 45x
According to question
$\because$ 60% = 45x Marks
$\therefore$ Maximum Marks (100%) = $\frac{45x}{60}\times 100$ Marks
= 75x Marks

$\therefore$ Maximum marks of each paper = $\frac{75x}{6}$
$\therefore$ 60% of marks each paper = $\frac{75x}{6}\times\frac{60}{100}$
$\Rightarrow$ 7.5 x
Hence, The number of papers which he scored less than 60% the maximum marks = 5x, 6x, 7x
$\therefore$ Required numbers = 3

68. A bank employee drives 10 km towards South from her house and turns to her left and drives another 20 km. She again turns left and drives 40 km, then she turns to her right and drives for another 5 km. She again turns to her right and drives another 30 km to reach her bank where she works. What is the shortest distance between her bank and her house?
(a) 20 km (b) 25 km
(c) 30 km (d) 35 km

Ans. (b) :



The shortest distance between her bank and her house = 25 km

69. Integers are listed from 700 to 1000. In how many integers is the sum of the digits 10?
(a) 6 (b) 7
(c) 8 (d) 9

Ans. (d) :
Rqeuired Numbers {703, 730, 712, 721; 802, 820, 811; 901, 910}
Hence, Total numbers = 9

70. A boy plays with a ball and he drops it from a height of 1.5 m. Every time the ball hits the ground, it bounces back to attain a height 4/5[th] of the previous height. The ball does not bounce further if the previous height is less than 50 cm. What is the number of times the ball hits the ground before the ball stops bouncing?
(a) 4 (b) 5
(c) 6 (d) 7

Ans. (b) : $\because$ Initial height = 1.5 m $\Rightarrow$ 150 cm
The height after first Bounce = $150\times\frac{4}{5}\Rightarrow 120$cm
The height after second Bounce = $120\times\frac{4}{5}\Rightarrow 96$cm
The height after third Bounce = $96\times\frac{4}{5}\Rightarrow 76.8$cm
The height after fourth Bounce = $76.8\times\frac{4}{5}\Rightarrow 61.44$cm
The height after fifth Bounce = $61.44\times\frac{4}{5}\Rightarrow 49.152$cm
$\because$ The ball does not bounce further if the previsous height is less than 50 cm.
$\therefore$ The number of times the ball hits the ground before the ball stop bouncing = 5

71. A biology class of high school predicted that a local population of animals will double in size every 12 years. The population at the beginning of the year 2021 was estimated to be 50 animals. If P represents the population after n years, then which one of the following equations represents the model of the class for the population?
(a) $P = 12 + 50n$ (b) $P = 50 + 12n$
(c) $P = 50\,(2)^{12n}$ (d) $P = 50\,(2)^{n/12}$

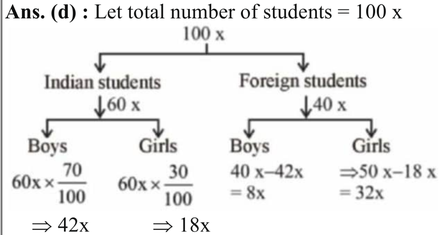
Ans. (d) : Amount = Principal $\left(1+\frac{R}{100}\right)^n$

$$2x = x\left(1+\frac{R}{100}\right)^{12}$$
$$2 = \left(1+\frac{R}{100}\right)^{12}$$
$$2^{\frac{1}{12}} = \left(1+\frac{R}{100}\right)$$
$$\left(1+\frac{R}{100}\right) = 2^{\frac{1}{12}}$$

Here,
$$P = 50\left(1+\frac{R}{100}\right)^n$$
$$P = 50\left[(2)^{\frac{1}{12}}\right]^n$$
$$\boxed{P = 50(2)^{\frac{n}{12}}} \qquad \left\{\because \left(1+\frac{R}{100}\right) = 2^{\frac{1}{12}}\right\}$$

72. In a class, 60% of students are from India and 50% of the students are girls. If 30% of the Indian students are girls, then what percentage of foreign students are boys?
(a) 45% (b) 40%
(c) 30% (d) 20%

Ans. (d) : Let total number of students = 100 x



($\because$ 50% Girls of the students)
Required % = $\frac{8x}{40x}\times 100\% = 20\%$